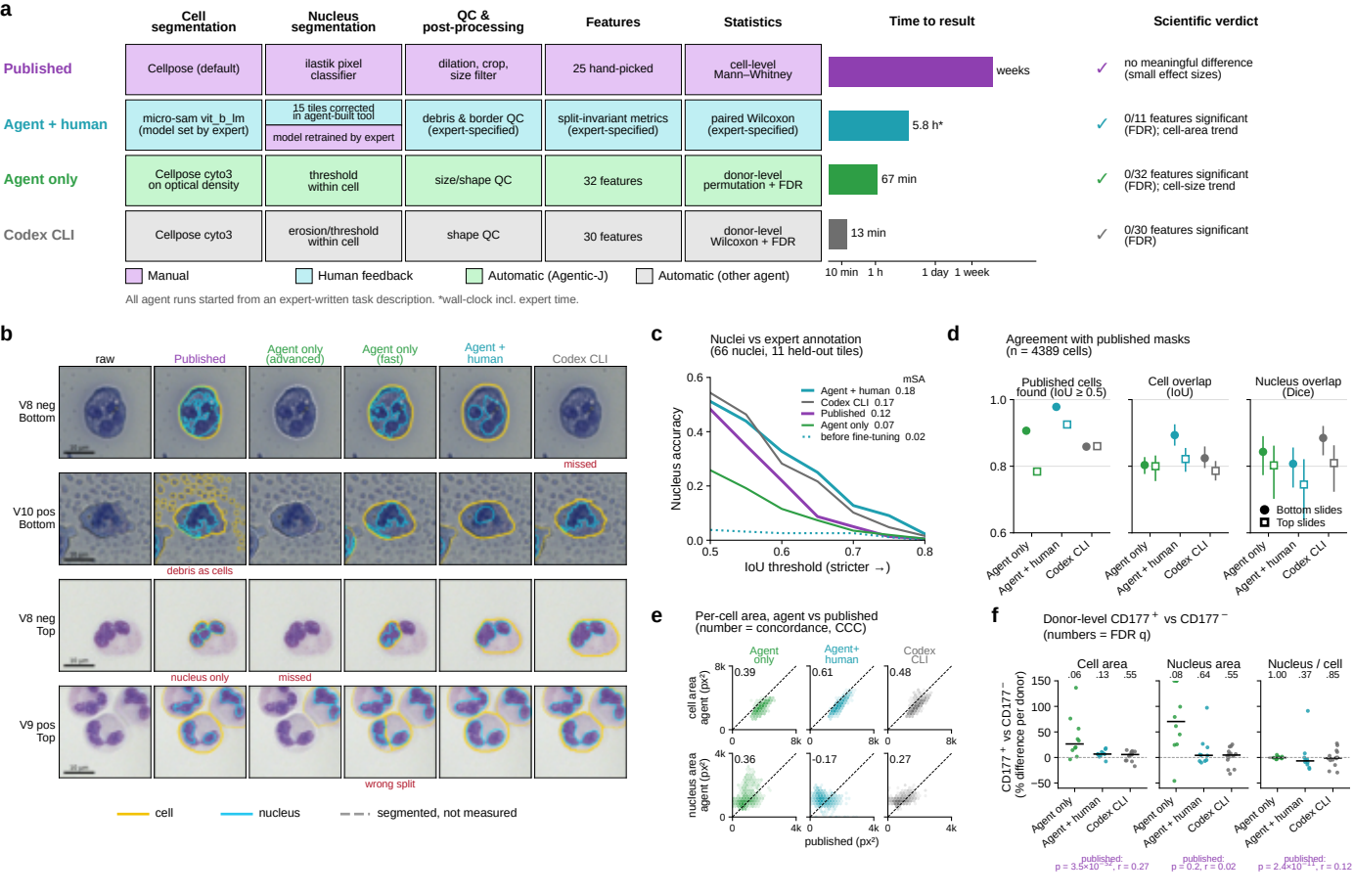

a
Cell segmentation
Nucleus segmentation
QC & post-processing
Features
Statistics
Time to result
Scientific verdict
Published
Cellpose (default)
ilastik pixel classifier
dilation, crop, size filter
25 hand-picked
cell-level Mann–Whitney
weeks
no meaningful difference (small effect sizes)
Agent + human
micro-sam vit_b_lm (model set by expert)
15 tiles corrected in agent-built tool
model retrained by expert
debris & border QC (expert-specified)
split-invariant metrics (expert-specified)
paired Wilcoxon (expert-specified)
5.8 h*
0/11 features significant (FDR); cell-area trend
Agent only
Cellpose cyto3 on optical density
threshold within cell
size/shape QC
32 features
donor-level permutation + FDR
67 min
0/32 features significant (FDR); cell-size trend
Codex CLI
Cellpose cyto3
erosion/threshold within cell
shape QC
30 features
donor-level Wilcoxon + FDR
13 min
0/30 features significant (FDR)
10 min
1 h
1 day
1 week
Manual
Human feedback
Automatic (Agentic-J)
Automatic (other agent)
All agent runs started from an expert-written task description. *wall-clock incl. expert time.
b
raw
Published
Agent only (advanced)
Agent only (fast)
Agent + human
Codex CLI
V8 neg Bottom
missed
V10 pos Bottom
debris as cells
V8 neg Top
nucleus only
missed
V9 pos Top
wrong split
cell
nucleus
segmented, not measured
c
Nuclei vs expert annotation
(66 nuclei, 11 held-out tiles)
mSA
Agent + human 0.18
Codex CLI 0.17
Published 0.12
Agent only 0.07
before fine-tuning 0.02
Nucleus accuracy
IoU threshold (stricter →)
d
Agreement with published masks
(n = 4389 cells)
Published cells found (IoU ≥ 0.5)
Cell overlap (IoU)
Nucleus overlap (Dice)
Bottom slides
Top slides
Agent only
Agent + human
Codex CLI
e
Per-cell area, agent vs published
(number = concordance, CCC)
Agent only
Agent+ human
Codex CLI
cell area agent (px²)
nucleus area agent (px²)
0.39
0.61
0.48
0.36
-0.17
0.27
published (px²)
f
Donor-level CD177⁺ vs CD177⁻
(numbers = FDR q)
Cell area
Nucleus area
Nucleus / cell
.06
.13
.55
.08
.64
.55
1.00
.37
.85
CD177⁺ vs CD177⁻ (% difference per donor)
Agent only
Agent + human
Codex CLI
published: p = 3.5×10⁻⁵⁴, r = 0.27
published: p = 0.2, r = 0.02
published: p = 2.4×10⁻¹¹, r = 0.12